

Q1.

A uniform solid is formed by rotating the region enclosed by the positive x -axis, the line $x = 2$ and the curve $y = \frac{1}{2}x^2$ through 360° around the x -axis.

(a) Find the centre of mass of this solid.

(5)

(b) The solid is placed with its plane face on a rough inclined plane and does not slide.

The angle between the inclined plane and the horizontal is gradually increased.

When the angle between the inclined plane and the horizontal is α , the solid is on the point of toppling.

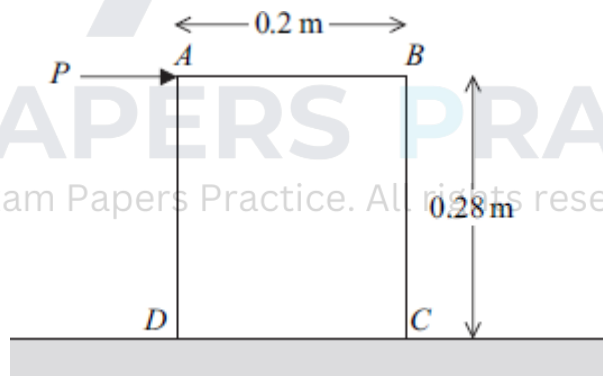
Find α , giving your answer to the nearest 0.1°

(2)

(Total 7 marks)

Q2.

A box of breakfast cereal, of total mass 0.4 kg , rests on a rough horizontal plane. It can be assumed that the centre of the mass of the box with its contents is on a vertical line through the centre of the box. The diagram shows a vertical cross section $ABCD$ through the centre of mass of the box and its contents, where $AB = 0.2 \text{ m}$ and $BC = 0.28 \text{ m}$. A horizontal force of magnitude P is applied at A , in the direction of AB .

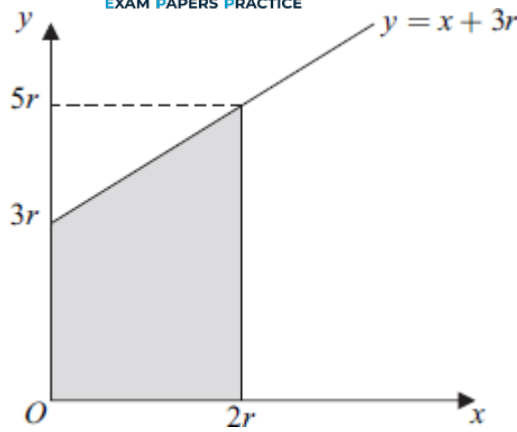


The coefficient of friction between the box and the plane is μ . As P is gradually increased from zero, the box slides before it topples if, and only if, $\mu < k$. Find the value of k .

(Total 6 marks)

Q3.

The region bounded by the line $y = x + 3r$, the y -axis, the x -axis and the line $x = 2r$ is shown in the diagram.



This region is rotated about the x -axis to form a frustum, of volume $\frac{98\pi r^3}{3}$, of a uniform solid cone.

- (a) Using integration, find the distance of the centre of mass of the frustum of the cone from O . (6)
- (b) The frustum of a uniform solid cone with radii $3r$ and $5r$ and height $2r$ has weight W . The frustum stands on a horizontal surface. The diagram shows a cross-section, $ABCD$, which includes the axis of symmetry, of this frustum. A force P is applied to the point B and acts in the same plane as this cross-section at an angle of θ below the horizontal.



The frustum does not slide and is about to topple about A .

- (i) Show that

$$P = \frac{3W}{2(\cos \theta + \sin \theta)}$$

(4)

- (ii) Find, in terms of W , the minimum possible value of P .

(3)

- (iii) State the value of θ for which P is a minimum.

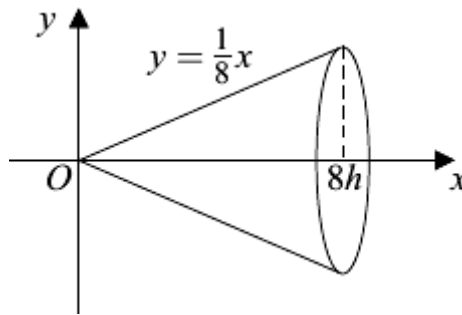
(1)

(Total 14 marks)

Q4.

A uniform solid cone is formed by rotating the finite region bounded by the lines

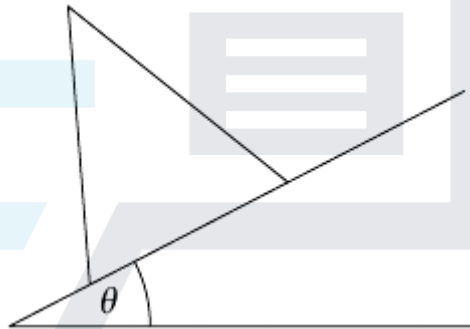
$y = \frac{1}{8}x$, $y = 0$ and $x = 8h$ through 2π radians about the x -axis. The cone is shown in the diagram.



- (a) Use integration to show that the distance of the centre of mass of the cone from the origin is $6h$.

(5)

- (b) The cone rests in equilibrium with its plane face on a rough plane inclined at an angle θ to the horizontal, as shown in the diagram.



- (i) On the diagram, show all the forces acting on the cone.

(2)

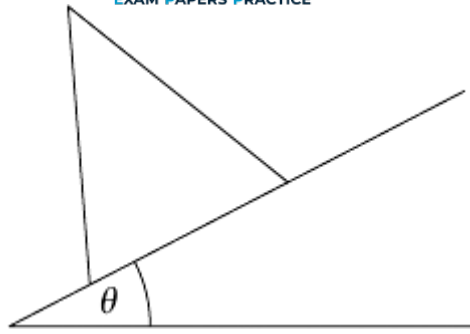
- (ii) If the plane is sufficiently rough to prevent slipping, find the maximum value of θ for the cone to remain in equilibrium without toppling.

(3)

- (c) The cone is now placed with its plane face on another rough inclined plane. The coefficient

of friction between the cone and this plane is $\frac{6}{13}$. The angle α between the inclined plane and the horizontal is gradually increased from $\alpha = 0^\circ$.

Determine whether the cone will topple first or slide first.

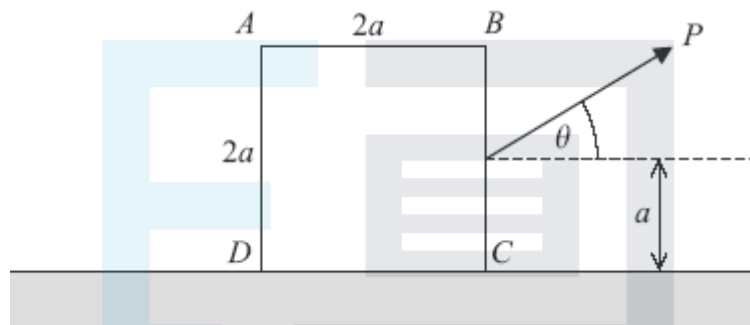


(4)

(Total 14 marks)

Q5.

A uniform cube, of side $2a$ and mass m , rests on a rough horizontal plane. The diagram shows a vertical cross-section $ABCD$ through the centre of mass of the cube.



A force, of magnitude P , is applied at the mid-point of BC . This force acts in the plane $ABCD$ and makes an angle θ with the horizontal. The coefficient of friction between the cube and the plane is μ .

- (a) In the case where the cube does not slide but is on the point of toppling about the edge through C , find an expression for P in terms of m , g and θ .

(3)

©2025 Exam Papers Practice. All rights reserved.

- (b) In the case where the cube remains upright but is on the point of sliding along the plane, show that $P = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$.

(4)

- (c) Find an inequality that μ must satisfy if the cube slides before it topples.

(3)

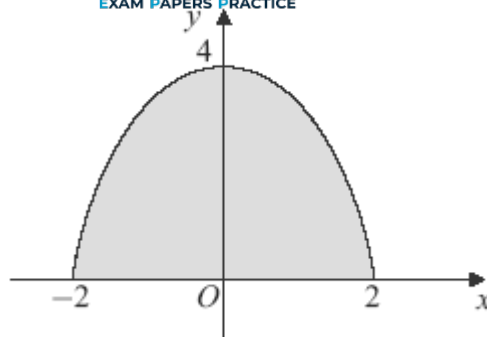
- (d) Would your answer in part (c) change if the mass of the cube were doubled? Explain why.

(2)

(Total 12 marks)

Q6.

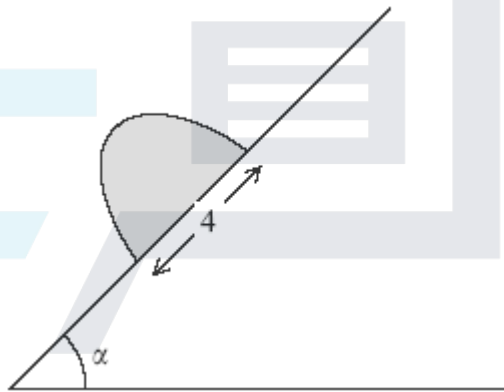
- (a) A uniform lamina is bounded by the curve $y = 4 - x^2$ and the x -axis, as shown in the diagram.



Given that the area of the lamina is $\frac{32}{3}$ square units, find the y-coordinate of the centre of mass of the lamina.

(5)

- (b) The cross-section of a uniform prism is the same shape as the lamina in part (a). The prism is placed on a plane inclined at an angle α to the horizontal with the rectangular base of the prism in contact with the inclined plane, as shown in the diagram.



Given that the prism is just about to topple and that no slipping occurs, find the value of α , giving your answer to the nearest degree.

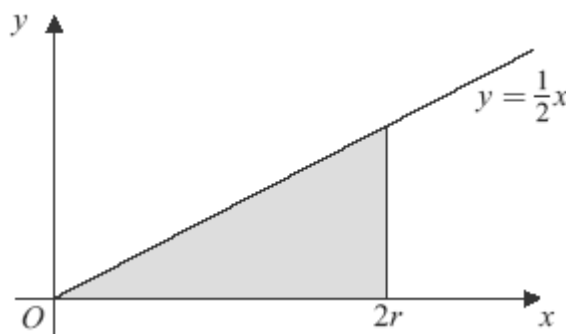
©2025 Exam Papers Practice. All rights reserved.

(4)

(Total 9 marks)

Q7.

The region bounded by the line $y = \frac{1}{2}x$, the x-axis and the line $x = 2r$ is shown in the diagram.

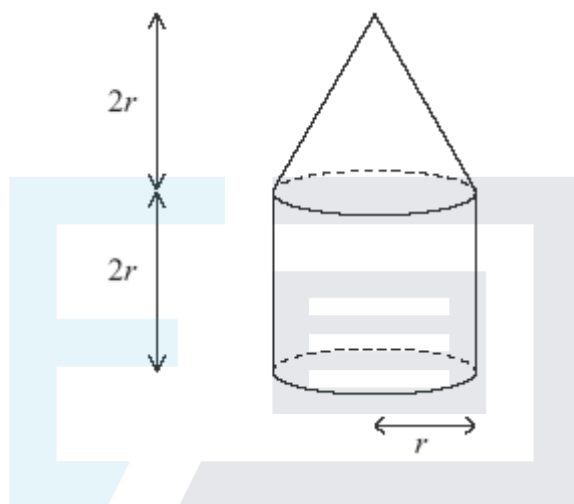


This region is rotated about the x -axis to form a uniform solid cone of height $2r$ and radius r .

- (a) Show, using integration, that the centre of mass of the cone is at a distance of $\frac{3r}{2}$ from the origin.

(5)

- (b) A rocket consists of two parts. The lower part of the rocket may be modelled as a uniform solid cylinder with radius r , height $2r$ and density ρ . The upper part of the rocket may be modelled as a uniform solid cone of radius r , height $2r$ and density $k\rho$, as shown in the diagram.

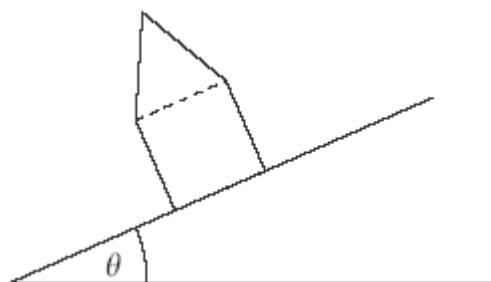


- (i) Show that the centre of mass of the rocket is at a distance of $\left(\frac{6+5k}{6+2k}\right)r$ from the base of the rocket.

(5)

- (ii) The rocket is now placed on a rough plane, which is inclined at an angle of θ to the horizontal, where $\tan \theta = \frac{2}{3}$.

©2025 Exam Papers Practice. All rights reserved.



Given that the rocket does **not** slide and is just on the point of toppling, find the value of k .

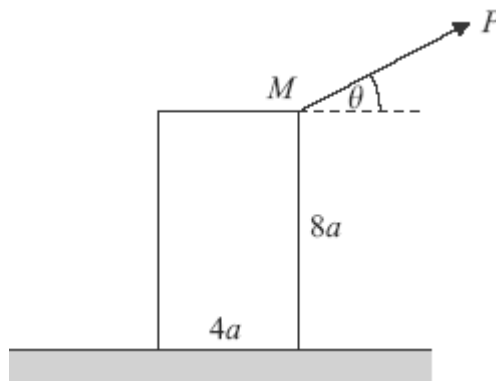
(5)

(Total 15 marks)

Q8.

A uniform block in the shape of a cuboid has weight W , a square base of side $4a$, a height

of $8a$ and stands on a rough horizontal surface. The coefficient of friction between the block and the surface is μ . A rope is attached to the point M , the midpoint of a top edge of the block.



The rope is pulled with a force P , which acts at an angle of θ above the horizontal and is perpendicular to the top edge.

- (a) Find P , in terms of W and θ , if the block is on the point of toppling.

(4)

- (b) Show that $P = \frac{\mu W}{\cos \theta + \mu \sin \theta}$ if the block is on the point of sliding.

(7)

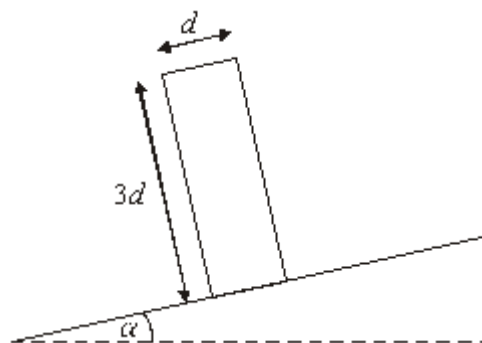
- (c) Given that $\tan \theta = 1$, find an inequality that μ must satisfy if the block slides before it topples.

(5)

(Total 16 marks)

Q9.

A uniform solid circular cylinder is in equilibrium with one plane face on a rough inclined plane. The plane is inclined to the horizontal at an angle α degrees, which can be varied. The cylinder has weight W , diameter d and height $3d$.



- (a) Draw a diagram showing the forces acting on the cylinder.

(2)

- (b) If the plane is sufficiently rough to prevent sliding, find the maximum value of α for the cylinder to remain in equilibrium.

(3)

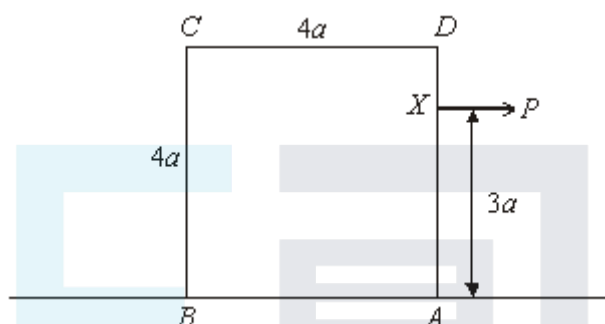
- (c) The coefficient of friction between the cylinder and the plane is $\frac{2}{9}$. If the value of α is gradually increased from zero, show that the cylinder will slide before it topples.

(5)

(Total 10 marks)

Q10.

A uniform cube, of side $4a$ and weight W , rests on a horizontal rough plane surface. The diagram shows a vertical cross-section $ABCD$ parallel to opposite faces of the cube and through its centre of mass.



A horizontal force P is applied to AD at a point X where $AX = 3a$. The coefficient of friction between the cube and the plane is μ .

- (a) In the case where the cube remains upright but slides along the plane under the action of P , determine an inequality for P in terms of W and μ .

(3)

- (b) In the case where the cube does not slide under the action of P but is on the point of toppling about the edge through A :

- (i) draw a diagram showing the lines of action of the forces on the cube;

(1)

- (ii) determine the value of P in terms of W .

(2)

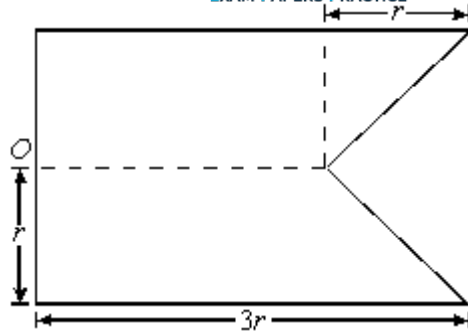
- (c) The value of P is gradually increased from zero. If $\mu = 0.6$, determine whether the cube will slide before it topples.

(3)

(Total 9 marks)

Q11.

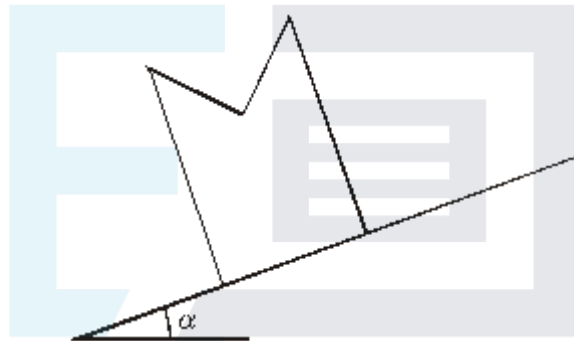
A solid, uniform cylinder, of radius r and length $3r$, has a cone of radius r and height r drilled out of one end. The diagram shows a cross section along the axis of symmetry of the body. Point O is the centre of the plane circular surface of the body.



- (a) Show that the centre of mass of the body is $\frac{43r}{32}$ from point O.

(6)

- (b) The body is placed with its plane surface on an adjustable rough slope as shown. The angle α which the plane makes with the horizontal is gradually increased from zero.



Assuming that the body does **not** slide, find, to the nearest degree, the value of angle α when the body just begins to topple.

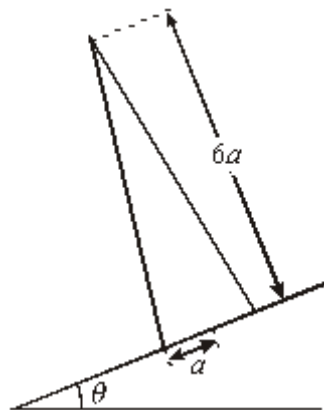
(3)

(Total 9 marks)

Q12.

©2025 Exam Papers Practice. All rights reserved.

A uniform solid cone of weight W rests in equilibrium with its plane face on a rough plane inclined at an angle θ to the horizontal. The cone is of height $6a$ and the base radius a .



- (a) Draw a diagram showing the forces acting on the cone.

(2)

- (b) The angle θ is gradually increased from $\theta = 0^\circ$.

Assuming that the plane is sufficiently rough to prevent slipping, show that the cone is about to topple when $\tan \theta = \frac{2}{3}$.

(3)

- (c) The cone is now placed on another rough inclined plane where the coefficient of friction between the cone and the plane is 0.6. The angle θ between the plane and the horizontal is gradually increased from $\theta = 0^\circ$. Determine whether the cone will topple or slide first.

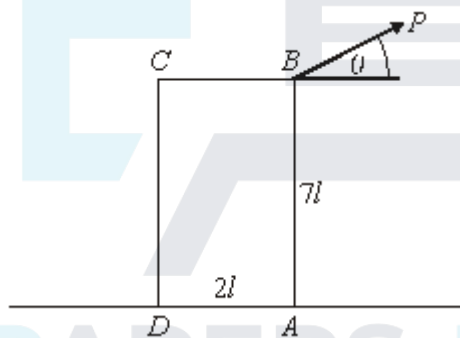
(6)

(Total 11 marks)

Q13.

A uniform solid cuboid of mass M is placed on a rough horizontal floor. The cuboid has a square base of side $2l$ and a height of $7l$.

A force, P , which is gradually increasing, is applied to the midpoint of, and perpendicular to, a top edge.



This force acts as shown in the diagram, where $ABCD$ is a vertical cross-section through the centre of mass of the cuboid.

The force P makes an angle θ with the horizontal.

The coefficient of friction between the block and the rough horizontal floor is $\frac{1}{5}$.

- (a) Show that, if the block is on the point of sliding,

$$P = \frac{Mg}{5\cos\theta + \sin\theta}$$

(6)

- (b) Find P when the block is on the point of toppling.

(3)

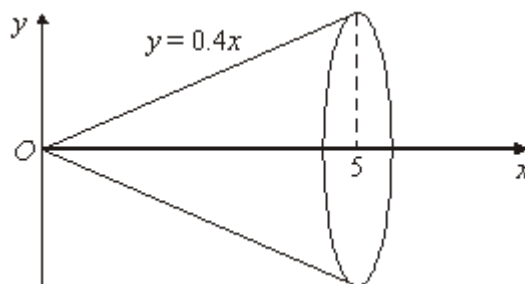
- (c) Find the range of possible values of $\tan \theta$ if the block topples before it slides.

(5)

(Total 14 marks)

Q14.

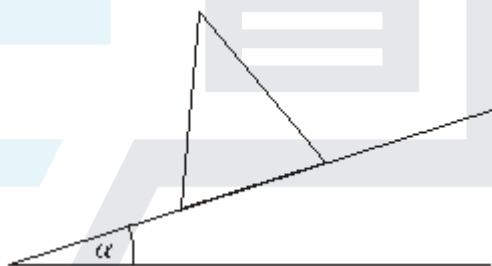
A uniform **solid** cone is formed by rotating the finite region bounded by the lines with equations $y = 0.4x$, $y = 0$ and $x = 5$ through 360° about the x -axis. The cone is shown in the diagram below.



- (a) Use integration to show that the distance of the centre of mass of the cone from the origin is $\frac{15}{4}$.

(5)

- (b) The cone is placed on a rough slope, which is at an angle α to the horizontal, as shown in the diagram.



The angle α is gradually increased.

Assuming that the cone does not slide, find the value of α when the cone is on the point of toppling.

(5)

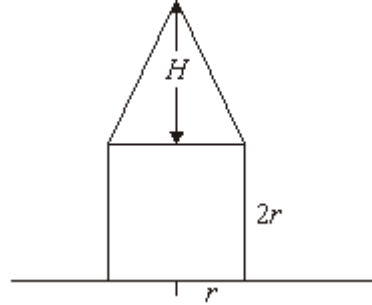
(Total 10 marks)

Q15.

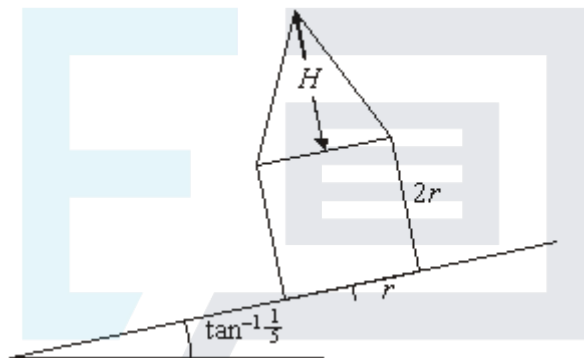
The diagram shows a cross-section of a child's toy, which consists of a uniform solid cylinder and a uniform solid right circular cone. One plane face of the cylinder is fixed to the plane face of the cone.

The cylinder has density ρ , radius r and height $2r$.

The cone has density 3ρ , height H and radius r .



- (a) Show that the centre of mass of the toy is $\frac{8r^2 + 8rH + H^2}{4(2r + H)}$ from its base. (6)
- (b) The toy is now placed on a rough plane inclined at an angle of $\tan^{-1} \frac{1}{5}$ to the horizontal.



Show that the maximum value of H if the toy is not to topple is $(6 + 2\sqrt{17})r$.

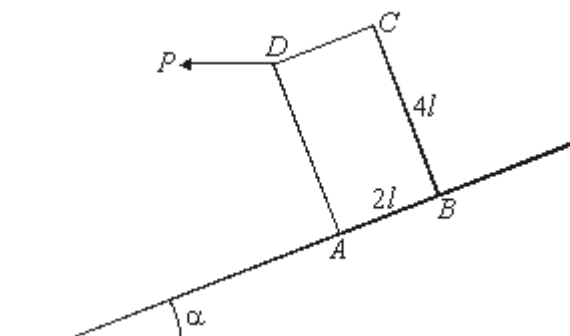
EXAM PAPERS PRACTICE (5)
(Total 11 marks)

Q16.

©2025 Exam Papers Practice. All rights reserved.

A uniform solid cuboid, of mass m , is placed on a rough inclined plane and remains at rest. cuboid has a square base of side $2l$ and a height of $4l$. A **horizontal** force P , which is gradually increasing, is then applied to the mid-point of, and perpendicular to, a top edge.

This force acts as shown in the diagram, where $ABCD$ is a vertical cross-section through the centre of mass of the cuboid.



The angle which the inclined plane makes with the horizontal is α . The coefficient of friction between the block and the plane is μ .

(a) Find P , if the block is on the point of toppling.

(4)

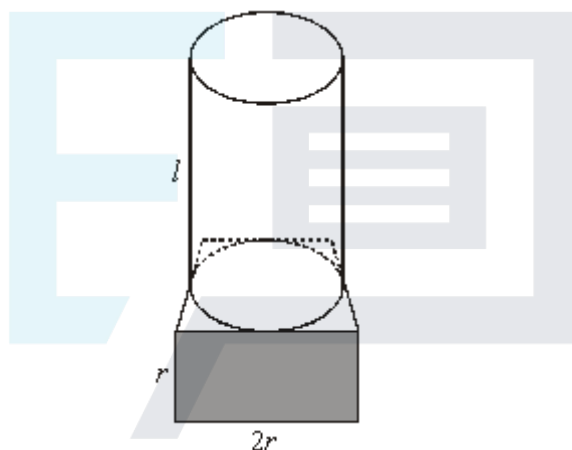
(b) Find P , if the block is on the point of sliding.

(7)

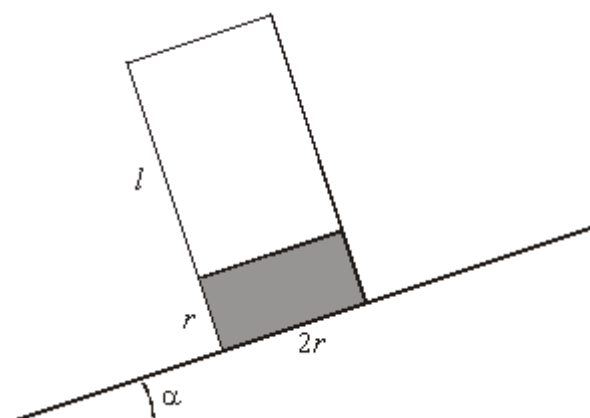
(Total 11 marks)

Q17.

A trophy may be modelled as a uniform solid cylinder, of radius r and length l , fixed to the top of a cuboid of square base of side $2r$ and height r . The density of the cylinder is ρ and the density of the cuboid is 6ρ . One of the plane faces of the cylinder is fixed to one of the square faces of the cuboid so that the centres of the two faces are coincident.



The trophy is placed on a plane inclined to the horizontal at an angle α , where $\tan \alpha = \frac{1}{4}$, so that a square base of the cuboid is in contact with the inclined plane.



If the trophy is on the point of toppling, show that

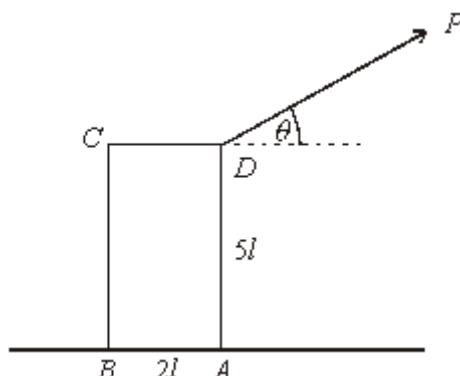
$$\pi l^2 - 6\pi r l - 168r^2 = 0$$

(Total 9 marks)

Q18.

A uniform solid cuboid of mass M is placed on a rough horizontal floor. The cuboid has a square base of side $2l$ and a height of $5l$.

A force, P , which is gradually increasing, is applied to the mid-point of, and perpendicular to, a top edge.



This force acts as shown in the diagram, where $ABCD$ is a vertical cross-section through the centre of mass of the cuboid.

The force, P , makes an angle θ with the horizontal. The coefficient of friction between the block and the rough horizontal floor is μ .

- (a) Show that the block is on the point of sliding when

$$P = \frac{\mu Mg}{\cos \theta + \mu \sin \theta}$$

(6)

- (b) Find P when the block is on the point of toppling.

(3)

©2025 Exam Papers Practice. All rights reserved.

- (c) Given that $\tan q = \frac{1}{7}$, find an inequality that μ must satisfy if the block slides before it topples.

(5)

(Total 14 marks)